

Answer - Key

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Q1) Data structure is a way of organising and storing data in the computer.

Q2) The time complexity is $O(\log n)$

Q3) Array.

Q4) Last in First Out.

Q5)

<u>Stack</u>	<u>Queue</u>
(i) It uses LIFO LIFO	(i) It uses FIFO.
(ii) It has pop/push functions.	(ii) It has enqueue/dequeue functions.

Q6) Recursion is method to write algorithm where the function calls itself until base condition is not met.

Eg. Fibonacci Algo.

Q7) Time Complexity is a measure of time.

Space Complexity is a measure of space.

Q8) ~~in~~ Linear Search is a algorithm where traverse the array from start to end and check for target continuously.

Algorithm :-

```
int target = 8;
int array = [1, 5, 8]
```

```
for (i=0; i < size(array); i++) {
```

```
    if (array[i] == target) {
        return 1;
    }
```

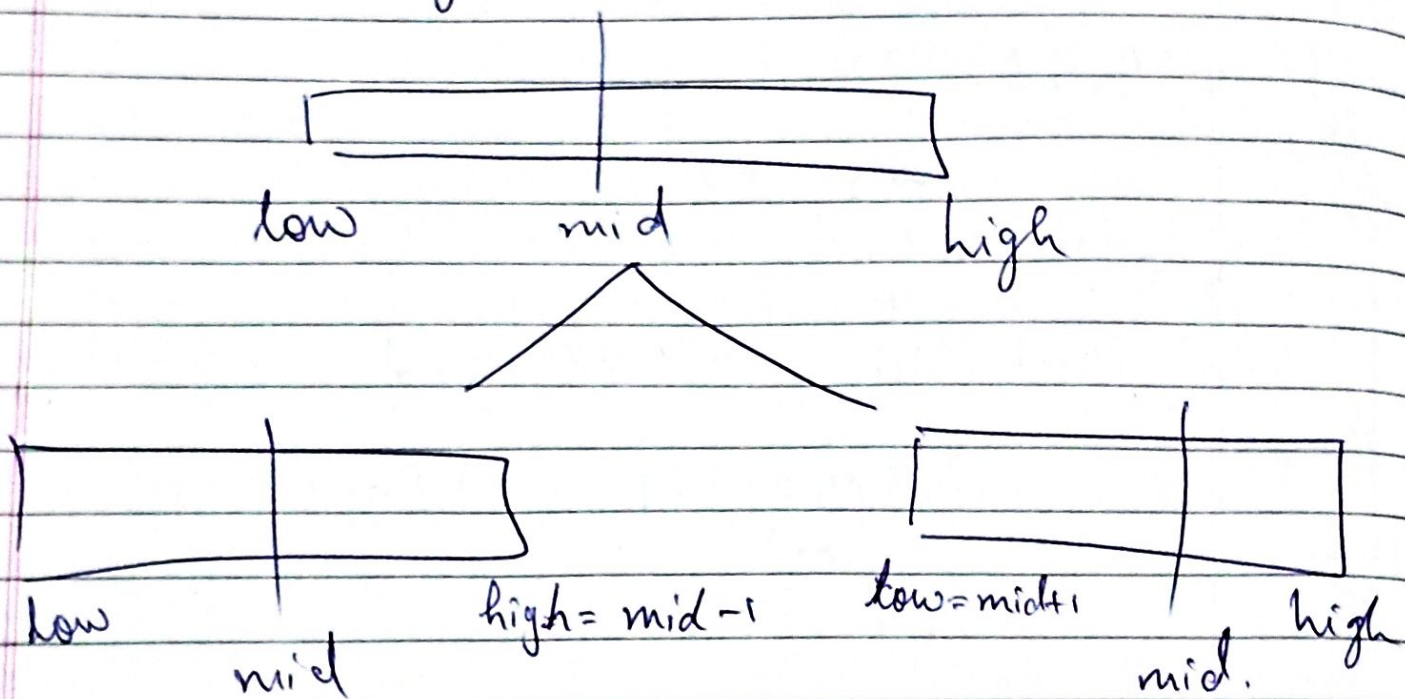
```
    }
    return 0;
```

Time Complexity = $O(n)$

Q9) ~~Time~~ Time Complexity = $O(n)$

Please give me marks.

Q10) Binary Search is a divide and conquer algorithm which splits the array every time and searches in one of the halves only.



Also, the array must be sorted.

The time Complexity is $O(\log n)$